

ORKHON, Mongolia

WMO: 442420

Lat: 49.1436N

Lon: 105.4022E

Elev: 7156

StdP: 11.27

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.8	-31.1	-40.6	0.7	-34.3	-36.8	0.9	-30.8	19.2	9.2	15.9	4.6	1.2	110	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.2	89.5	61.8	85.7	61.2	82.2	60.6	67.6	80.3	65.5	77.6	63.8	75.7	6.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	116.2	74.6	61.7	107.9	71.9	59.9	101.1	70.1	36.6	80.3	34.7	77.2	33.2	76.0	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	20.9	17.8	15.0	DB	-38.4	96.0	5.1	4.1	-42.1	98.9	-45.0	101.3	-47.9	103.6	-51.6	106.6
(5)				WB	-38.6	70.0	4.9	2.5	-42.1	71.8	-45.0	73.2	-47.7	74.6	-51.3	76.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	32.4	-13.4	-4.3	20.9	40.4	52.2	64.0	69.0	63.8	52.3	35.8	12.5	-6.9	(6)
(7)		DBStd	30.03	9.65	12.12	13.03	8.98	8.85	7.34	5.59	5.99	7.95	8.53	12.97	11.14	(7)
(8)		HDD50	8181	1965	1522	902	308	81	2	0	1	68	442	1125	1765	(8)
(9)		HDD65	12193	2430	1942	1367	739	404	103	21	93	385	904	1575	2230	(9)
(10)		CDD50	1748	0	0	0	19	150	422	588	430	136	3	0	0	(10)
(11)		CDD65	288	0	0	0	0	9	74	145	57	3	0	0	0	(11)
(12)		CDH74	3674	0	0	0	14	293	1064	1502	677	123	1	0	0	(12)
(13)		CDH80	1241	0	0	0	1	84	389	548	199	20	0	0	0	(13)
(14)	Wind	WSAvg	4.3	2.0	2.5	5.1	6.7	6.9	5.9	4.5	4.4	4.8	4.3	2.8	2.1	(14)
(15)	Precipitation	PrecAvg	11.8	0.1	0.1	0.2	0.3	0.9	2.2	3.1	3.0	1.2	0.4	0.2	0.2	(15)
(16)		PrecMax	17.1	0.2	0.2	0.6	0.7	2.6	5.6	5.2	7.4	2.3	1.7	0.6	0.5	(16)
(17)		PrecMin	8.2	0.0	0.0	0.0	0.1	0.1	1.0	1.1	1.4	0.3	0.0	0.0	0.0	(17)
(18)		PrecStd	2.2	0.1	0.1	0.2	0.2	0.6	1.1	1.3	1.4	0.6	0.4	0.1	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.0	31.8	59.4	75.8	88.5	94.6	95.2	90.8	83.1	68.8	49.5	26.1	(19)
(20)		0.4%	MCWB	16.4	25.9	41.1	49.3	57.7	60.7	65.5	62.6	58.2	48.1	37.9	21.4	(20)
(21)		2%	DB	12.3	25.0	51.7	68.4	81.8	88.5	89.9	85.2	77.4	61.4	42.0	18.9	(21)
(22)		2%	MCWB	10.3	21.0	37.8	45.6	54.3	59.5	64.1	62.6	56.6	45.1	32.5	15.8	(22)
(23)		5%	DB	6.4	19.6	45.6	63.1	76.7	84.4	86.0	81.2	72.8	56.1	35.8	13.5	(23)
(24)		5%	MCWB	4.8	16.7	34.5	43.7	52.8	59.0	63.5	62.0	54.6	42.0	29.5	11.4	(24)
(25)		10%	DB	1.5	14.1	40.4	57.7	71.5	80.1	82.1	77.2	68.1	51.2	30.8	9.0	(25)
(26)		10%	MCWB	0.3	11.9	31.3	41.4	50.3	58.1	62.8	61.2	52.7	39.5	26.5	7.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.7	25.8	42.6	50.5	60.7	66.4	70.8	69.0	61.2	48.7	37.9	21.9	(27)
(28)		0.4%	MCDWB	18.9	31.8	57.1	71.5	81.0	83.6	86.7	81.5	75.8	66.2	49.1	26.2	(28)
(29)		2%	WB	10.3	21.1	38.4	47.3	56.5	63.9	68.4	66.3	58.5	45.6	33.4	15.8	(29)
(30)		2%	MCDWB	12.1	25.0	50.7	65.1	77.4	79.3	81.8	78.6	73.0	60.4	41.1	18.5	(30)
(31)		5%	WB	4.8	16.9	34.8	44.7	53.6	62.2	66.7	64.1	56.1	42.4	29.6	11.5	(31)
(32)		5%	MCDWB	6.3	19.6	45.0	61.5	73.5	75.9	78.5	75.5	70.1	54.0	35.3	13.5	(32)
(33)		10%	WB	0.3	12.0	31.5	42.1	51.2	60.4	64.9	62.5	53.7	39.8	26.3	7.4	(33)
(34)		10%	MCDWB	1.5	14.0	39.9	56.7	69.6	75.1	76.4	72.9	66.3	50.3	31.1	8.9	(34)
(35)	Mean Daily Temperature Range		MDBR	17.9	23.5	25.4	25.8	27.2	25.2	22.2	22.5	26.1	25.1	20.5	16.9	(35)
(36)		5% DB	MCDBR	22.3	26.8	30.0	36.3	37.8	32.9	29.4	31.1	34.7	33.4	26.4	21.4	(36)
(37)			MCWBR	20.6	23.0	17.6	18.3	16.8	12.4	10.1	11.8	15.5	18.4	18.0	18.9	(37)
(38)			MCDBR	22.3	26.5	29.2	33.4	33.8	24.9	23.1	24.2	30.1	31.4	24.7	21.4	(38)
(39)		5% WB	MCWBR	20.6	22.8	17.7	17.8	16.1	10.5	10.1	10.9	13.9	17.7	17.3	18.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.204	0.219	0.250	0.282	0.308	0.318	0.311	0.295	0.261	0.240	0.220	0.206	(40)	
(41)		taud	2.406	2.402	2.443	2.450	2.398	2.449	2.506	2.538	2.593	2.584	2.483	2.398	(41)	
(42)		Ebn at Noon	289	307	311	308	302	298	299	301	305	297	279	272	(42)	
(43)		Edh at Noon	19	25	29	32	36	34	32	30	25	21	18	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	560	922	1379	1717	2004	2055	1926	1653	1386	923	572	431	(44)	
(45)		RadStd	23	42	55	79	117	123	110	108	73	49	28	27	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (7 neighbors)	N/A	N/A	-3.60	N/A	N/A	-1.06	N/A	N/A	N/A	N/A	

Nomenclature: See separate page